JOÃO VIANA

Data Scientist

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SUMMARY

During my PhD in theoretical physics, I spent most of my time developing theories and using computers to test them. Much of that is the same work data science asks for: building models, fitting them to data, and checking whether the fit means anything. I'm now moving that into data science and machine learning, an interest that has grown steadily over the past few years and that I wish to pursue.

PROJECTS

Impact of AI on Students

A synthetic Kaggle dataset about students AI usage. I clean out values left over from the data being synthetic. Built several regression models to predict end-of-semester GPA, and several classification models to flag burnout risk, tuned to catch 90% of high-risk students at the cost of accuracy.

Skills: Python (NumPy, Matplotlib, pandas, PyTorch, scikit-learn)

Anomaly detection on the MVTec AD database

Spotting defective products after training only on good images (zero-shot), tested on bottles, carpet, and hazelnuts. Convolutional autoencoders rebuild the image and flag bad rebuilds, better architectures are needed. PatchCore needs no training, compares each image to a memory bank of pretrained features, and was near perfect, and it also shows where the defect is.

Skills: Python (PyTorch, NumPy, Matplotlib), convolutional autoencoders, PatchCore, ROC/AUC.

RAG-Powered NumPy Documentation Assistant

A tiny 350M LLM given a search layer over the NumPy docs, to provide context before answering. Runs in three Docker containers (llm, backend, frontend) via Docker Compose. The plain model hallucinates functions and arguments. The version with RAG gets them right and admits when it can't.

Skills: Python, LLMs, RAG (ChromaDB), Ollama, FastAPI, Streamlit, Docker, Docker Compose, REST APIs.

SKILLS

- Maths: differential/integral calculus, linear algebra, statistics, multivariate analysis, complex analysis
- Coding: C++, Python (matplotlib, numpy, pandas, scipy, thread, beautifulsoup, scikit-learn, pytorch), R, git, unittests, CI/CD, Docker/Docker Compose, Ollama, REST API (FastAPI), ChromaDB
- Numerical Methods: ODE/PDE solver, minimisation algorithms, integration, Monte-Carlo integration
- OS: Linux, macOS, Windows
- Languages: Portuguese, English

RESEARCH

Controlling the pandemic during the SARS-CoV-2 vaccination rollout, Nature Communications 12, 3674 (2021), J. Viana, C. H. van Dorp, A. Nunes, et al..

Topic: International collaboration with the DGS/Ministry of Health to predict the number of COVID-19 hospitalisations and the impact of the vaccination rollout.

EDUCATION

PhD, Physics and Astronomy, F. Ciências ULisboa — 2022 - present

- P. Basler, L. Biermann, M. Mühlleitner, J. Müller, R. Santos, J. Viana — BSMPT v3: A Tool for Phase Transitions and Primordial Gravitational Waves in Extended Higgs Sectors, arXiv:2404.19037.
- M. Mühlleitner, J. Plotnikov, R. Santos, J. Viana — A Deep Dive into Baryon Asymmetry — the C2HDM, arXiv:2606.04229.

MSc, Nuclear and Particle Physics, F. Ciências ULisboa (Final grade: 19) — 2019 - 2022

- Thesis : Dark Matter, Gravitational Waves and Higgs invisible decays (Thesis grade: 20)

BSc, Physics, F. Ciências ULisboa (Final grade: 18) — 2016 - 2019